

Principal component Analysis:

Given:

$m \times n$ dataset

Problem:

$m \times K$ dataset where $K < n$

many original feats $\rightarrow$ few principal components

maximize variance $\leftrightarrow$ minimize reconstruction error

$$w_1 = \arg \max \text{Var}(Xw)$$

$$\text{s.t. } \|w\| = 1$$

$$X = m \times n$$

$$w_1 = n \times 1$$

$$Xw_1 = m \times 1$$



$$\text{Var}(X_c w) = \frac{1}{n-1} \left((X_c w)^T \cdot (X_c w) \right)$$

$$\text{var}(X_c w) = \frac{w^T \left(\overset{\text{co-variance matrix}}{X_c^T X_c} \right) w}{(n-1)} \quad \rightarrow \text{direction vector}$$

$$\max \frac{w^T (X_c^T X_c) w}{n-1}$$

$$\text{s.t. } w^T w = 1$$

$$X_c = m \times n$$

$$X_c^T = n \times m$$

$$X_c^T \cdot X_c = (n \times m) \times (m \times n)$$

$$= n \times n$$

for any matrix A,

$$A v = \lambda v$$

λ = eig. values

v = eig. vectors,

$$\text{let } X_c^T X_c = V_{n \times n}$$

$$V w = \lambda w$$

λ = eigen value

w = eigen vector.

w shape = $n \times 1$

properties:

Eigen vectors are orthogonal.

$$w_i^T w_j = 0 \quad i \neq j$$

$$w_i = n \times 1$$

$$w_j = n \times 1$$

Ordering the components:

$$\lambda_1 > \lambda_2 > \lambda_3 \dots > \lambda_n$$

$\downarrow$

$$w_1, w_2, w_3 \dots w_n$$

$$W = [w_1 \ w_2 \ w_3 \ \dots \ w_n] \rightarrow \text{all eigen vectors}$$

Choosing k eigen vectors, $k < n$

$$W_k = [w_1, w_2, w_3, \dots, w_k]$$

$$Z = X_c W_k \rightarrow n \times k$$

Diagram showing dimensions: $n \times k$ (circled) and $m \times n$ (circled) with arrows indicating the relationship between the matrices.

Reconstruct:

$$\hat{X}_c = Z \cdot W^T$$

$$\hat{X} = \hat{X}_c + \mu$$

$$\text{Reconstruction error} = \|X - \hat{X}\|_2$$

Explained variance ratio: (EVR)

$$EVR_k = \frac{\lambda_k}{\sum_{i=1}^n \lambda_i}$$

Steps for PCA

① Start with data matrix X with n columns

② Normalize $X_c = \frac{X - \mu}{\sigma}$

③ compute the covariance matrix $= \frac{1}{n-1} (X_c^T \cdot X_c)$

④ compute the eigenvalue and eigenvector of covariance matrix.

no. of eigen values $= n$, shape of eigenvectors $= n \times n$

- ⑤ sort the eigen values and correspondingly eigen vector.
- ⑥ select top k eigen values and corresponding eigen vectors (w_k)
- ⑦ $Z = X_G \cdot w_k$. Z is the transformed dataset.

for a ML model,

Training:

{ Data $\rightarrow$ split data $\rightarrow$ Scale train data $\rightarrow$ Fit PCA $\rightarrow$ train model

Validation/testing,

{ scale val/test using mean and variance from train data.
Transform using eigenvectors got from training pipeline
predict using trained model.